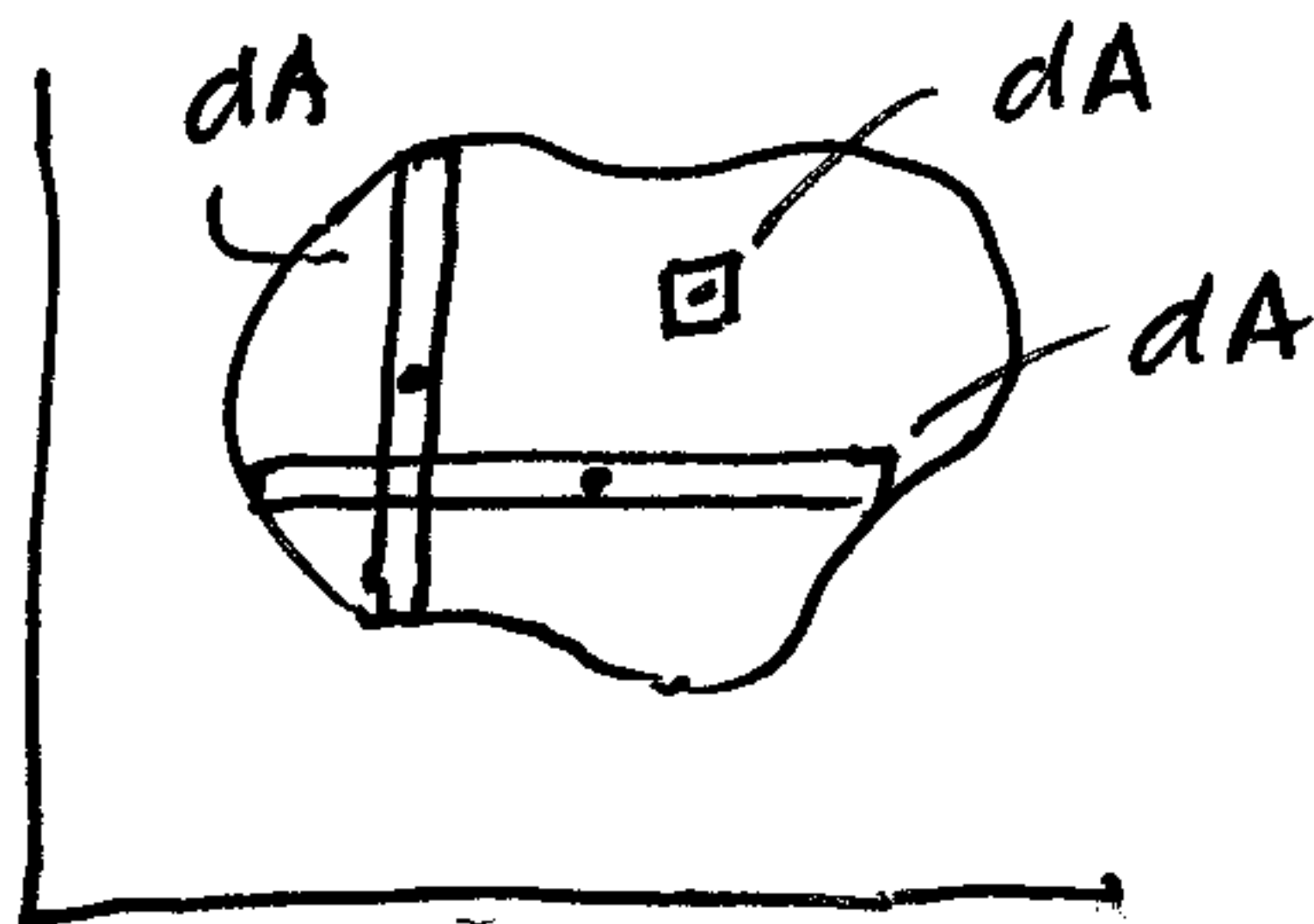


Centroids by Integration

Best choices of dA ?

Possible to use:



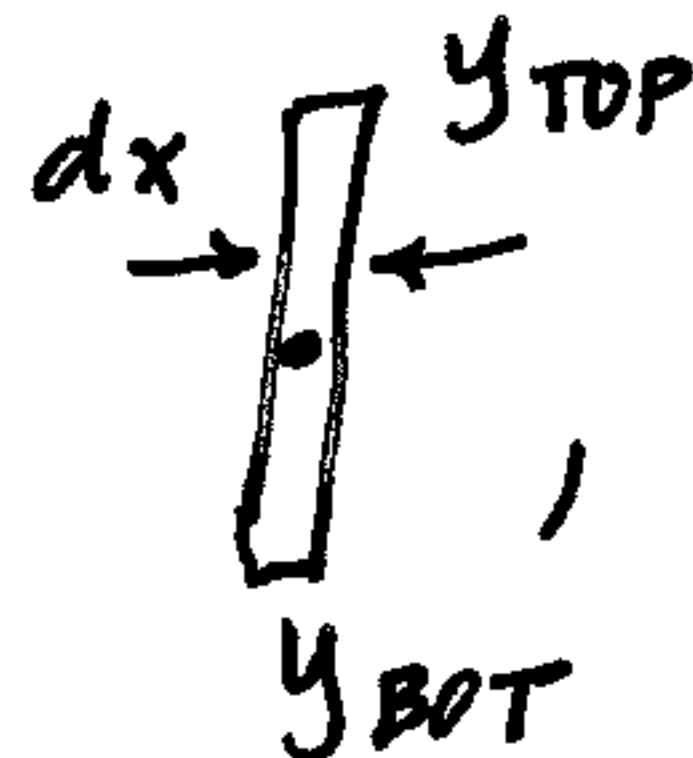
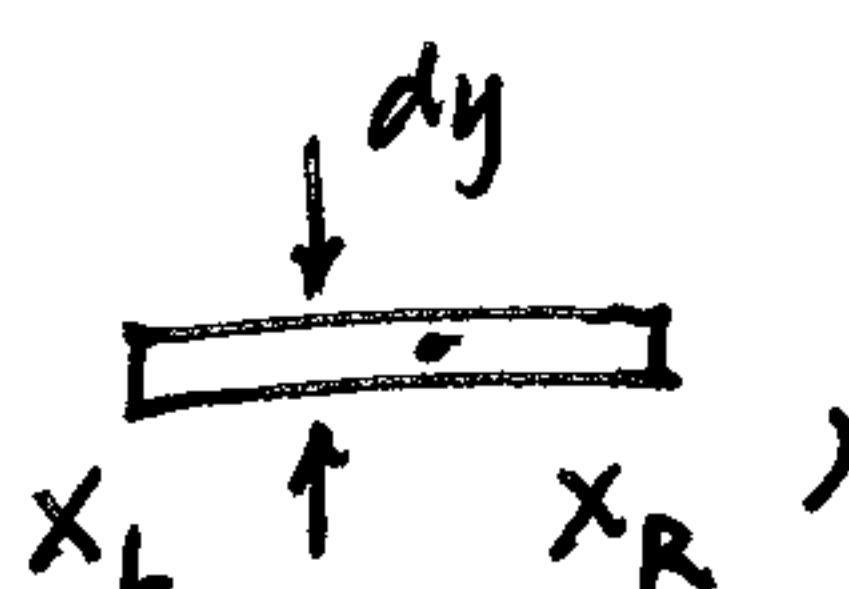
Since dA thickness $= dy$,
define x_L and x_R
in terms
of y .

$$\tilde{y} = y$$

$$\tilde{x} = \frac{(x_L + x_R)}{2}$$

$$dA = (x_R - x_L) \cdot dy$$

Single
Integral



Since dA thickness $= dx$,
define y_{TOP} + y_{BOT}
in terms
of x .

$$\tilde{x} = x$$

$$\tilde{y} = \frac{(y_{TOP} + y_{BOT})}{2}$$

$$dA = (y_{TOP} - y_{BOT}) \cdot dx$$

Double
Integral

$$\square \frac{dy}{dx}$$

$$\tilde{y} = y$$

$$\tilde{x} = x$$

$$dA = dx dy$$

or $dy dx$

To me,

- ① Double Integral is easiest.
 dA easy.

See above.

$\tilde{x}, \tilde{y}$ easy.

Also
easy for
Ch-10
Area
moments
of Inertia
(MOI's)

- ② Next easiest: (also for Ch 10 MOI's)

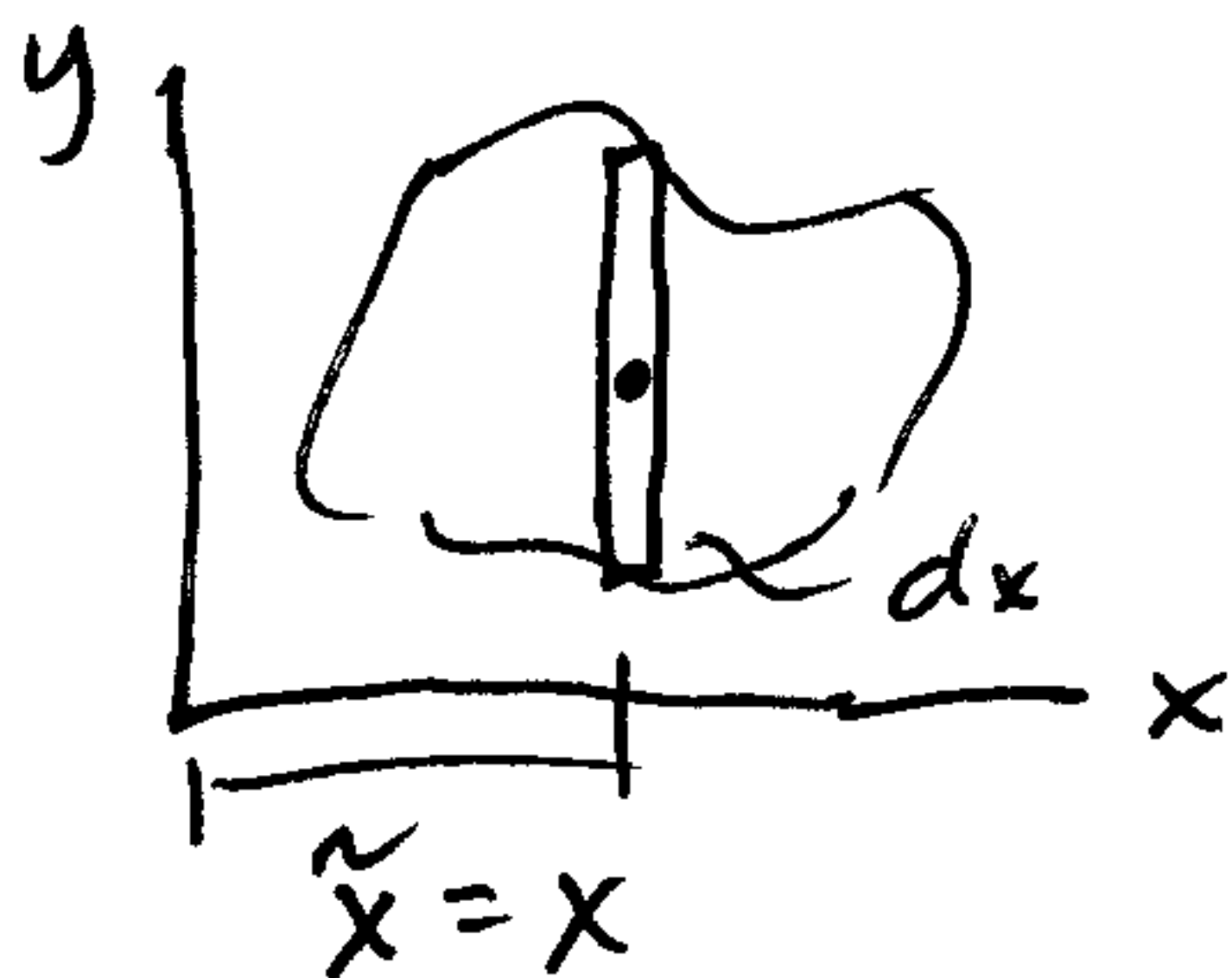
(a) For $\tilde{y}$, use element parallel to x axis:



Then, $\tilde{y} = y$.

$$dA = (x_R - x_L) \cdot dy$$

(b) For $\tilde{x}$, use element parallel to y axis:



Then, $\tilde{x} = x$

$$dA = (y_{TOP} - y_{BOT}) \cdot dx$$